



AQ7.1: Activity Questions 1 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

1 point

Choose the set of correct options.

☐

If a $3 \times 3 \times 3$ matrix AA is similar to the identity matrix of order 33, then AA must be the identity matrix of order 3.

☐

If a $3 \times 3 \times 3$ matrix AA is similar to a diagonal matrix of order 33, then AA must be a diagonal matrix of order 3.

☐

If AA and BB are similar matrices, then they are also equivalent matrices.

☐

If AA and BB are equivalent matrices, then they are also similar matrices.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If a $3 \times 3 \times 3$ matrix AA is similar to the identity matrix of order 33, then AA must be the identity matrix of order 3.

If AA and BB are similar matrices, then they are also equivalent matrices.

Consider the linear transformation given below:

$$T: R^3 \rightarrow R^2 \quad T: R^3 \rightarrow R^2$$

$$T(x, y, z) = (x - y, y + z) \quad T(x, y, z) = (x - y, y + z)$$

Consider two ordered bases $\beta_1 = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ and $\beta_2 = \{(1, 0, 1), (0, 1, 1), (1, 1, 0)\}$ of R^3 and consider the standard ordered basis $\gamma = \{(1, 0), (0, 1)\}$ of R^2 . Let AA and BB be the matrices corresponding to the linear transformation T with respect to the bases β_1 and β_2 for the domain, respectively and γ for the co-domain. Let QQ and PP be matrices such that $B = QAPB = QAP$.

Answer the questions 2,3, and 4 using the above information.

1 point

Which of the following matrices represents AA ?

☐

$$\begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

☐

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

☐

$$\begin{bmatrix} 1 & 1 \\ -1 & 1 \\ 0 & 1 \end{bmatrix}$$

☐


$$\begin{bmatrix} 1 & 1 \\ -1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 & 1 & 2 & 1 \end{bmatrix}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 & 1 & 0 & 1 \end{bmatrix}$$

1 point

Which of the following matrices can be PP ?

☐

There does not exist any matrix PP which satisfies the property $B = QAPB = QAP$.

☐

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 2 & 1 & 1 & 0 \end{bmatrix}$$

☐

$$\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 2 & 1 & -2 & 1 & 2 & 1 & -2 & 1 & 2 & 1 & 2 & 1 & -2 & 1 \end{bmatrix}$$

☐

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 \end{bmatrix}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 \end{bmatrix}$$

1 point

Which of the following statements is true for QQ ?

☐

QQ can be the identity matrix of order 3.

☐

QQ can be the identity matrix of order 2.

☐

Whatever the matrix PP we choose, QQ is always unique.

☐

QQ can be the matrix $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 & 0 \end{bmatrix}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

QQ can be the identity matrix of order 2.

QQ can be the matrix $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

1 point

Choose the set of correct options.

☐ If two square matrices of the same order have the same determinants, then they must be similar to each other.

☐ Similar matrices have the same rank.

☐

If AA and BB are similar matrices, and $Ax = b$ has a unique solution, then $Bx = b$ also has a unique solution.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Similar matrices have the same rank.

If AA and BB are similar matrices, and $Ax = b$ has a unique solution, then $Bx = b$ also has a unique solution.

1 point

Choose the set of correct options.

☐

If $\det A$ or $\det B$ is nonzero, then $ABAB$ and $BABA$ are similar.

☐

$A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$ are similar.

☐

$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 0 & 4 \end{bmatrix}$ are similar.

☐

$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ are not similar.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $\det A$ or $\det B$ is nonzero, then $ABAB$ and $BABA$ are similar.

$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ are not similar.

1 point

Which of the following options are correct?

☐ Two invertible matrices of same order are equivalent.

☐

$A = \begin{bmatrix} 17 & 6 \\ 21 & -1 \\ 52 & -3 \end{bmatrix}$ and $B = \begin{bmatrix} 23 & -2 \\ 11 & 4 \\ 12 & -6 \end{bmatrix}$ are equivalent.

☐

$A = \begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$ are equivalent.

☐

Consider two linear transformations $T_1(x, y, z) = (x + y - z, x + 4y - 2z, -2x + y + z)$ and $T_2(x, y, z) = (x + y, x - z, y + z)$. Let A and B denote the matrix representation of T_1 and T_2 with respect to the standard basis of $\mathbb{R}^3$. Then A and B are equivalent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Two invertible matrices of same order are equivalent.

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 4 & -2 \\ -2 & 1 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \\ 2 & 1 & 1 \end{bmatrix} \text{ are equivalent.}$$

Consider two linear transformations $T_1(x, y, z) = (x + y - z, x + 4y - 2z, -2x + y + z)$ and $T_2(x, y, z) = (x + y, x - z, y + z)$. Let A and B denote the matrix representation of T_1 and T_2 with respect to the standard basis of $\mathbb{R}^3$. Then A and B are equivalent.

Level 2

1 point

Let E be the set of 2×2 matrices which are similar to a scalar matrix $A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$.

Choose the set of correct statements.

☐

E is an infinite set.

☐

Cardinality of E is finite.

☐

E is singleton set.

☐

E consists of all the scalar matrices of order n .

No, the answer is incorrect.

Score: 0

Accepted Answers:

Cardinality of E is finite.

E is singleton set.

1 point

Which of the following option(s) is(are) true?

☐

If A and B are similar matrices then A^{-1} and B^{-1} are similar matrices.

☐

If A and B are similar matrices then A^2 and B^2 are similar matrices.

☐

If A^2 and B^2 are similar matrices then A and B are similar matrices.

☐

If A and B are similar matrices then A^T and B^T are similar matrices.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If A and B are similar matrices then A^{-1} and B^{-1} are similar matrices.

If A and B are similar matrices then A^2 and B^2 are similar matrices.

If A and B are similar matrices then A^T and B^T are similar matrices.

1 point

Let us consider the following matrices.

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad A = [1101], B = [1001], C = [1011]$$

Choose the set of correct options.

☐

AA and BB have the same rank, and same determinant, but they are not similar matrices.

☐

AA and CC have the same rank, and same determinant, but they are not similar matrices.

☐

BB and CC have the same rank, and same determinant, but they are not similar matrices.

☐ None of the options are true.

No, the answer is incorrect.

Score: 0

Accepted Answers:

AA and BB have the same rank, and same determinant, but they are not similar matrices.

BB and CC have the same rank, and same determinant, but they are not similar matrices.

[Check Answers](#)

Your score is: 0/10